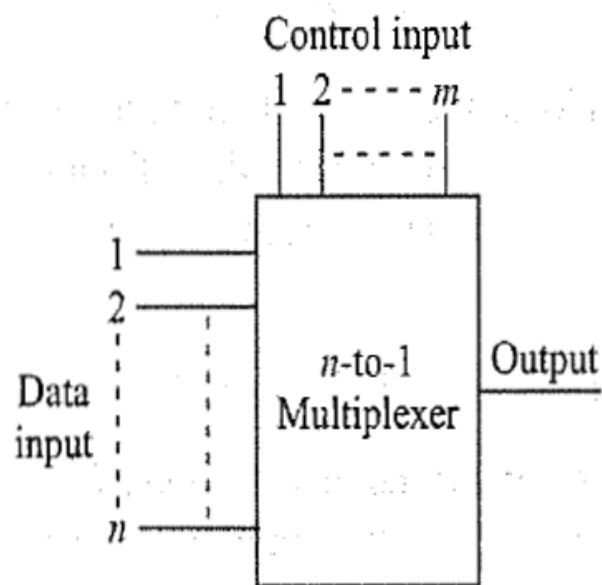


Multiplexers

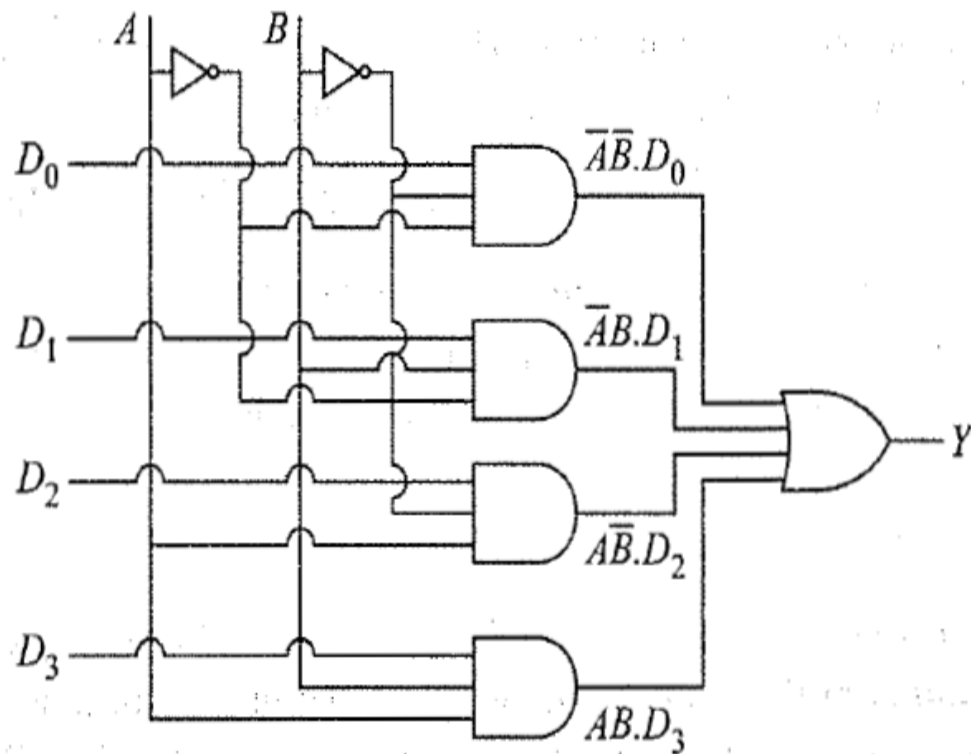
Multiplex means *many into one*. A *multiplexer* is a circuit with many inputs but only one output. By applying control signals, we can steer any input to the output. Thus it is also called a *data selector* and control inputs are termed select inputs. Figure 4.1a illustrates the general idea. The circuit has n input signals, m control signals and 1 output signal. Note that, m control signals can select at the most 2^m input signals thus $n \leq 2^m$.



(a)

A	B	Y
0	0	D_0
0	1	D_1
1	0	D_2
1	1	D_3

(b)



(c)

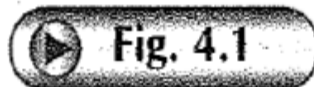
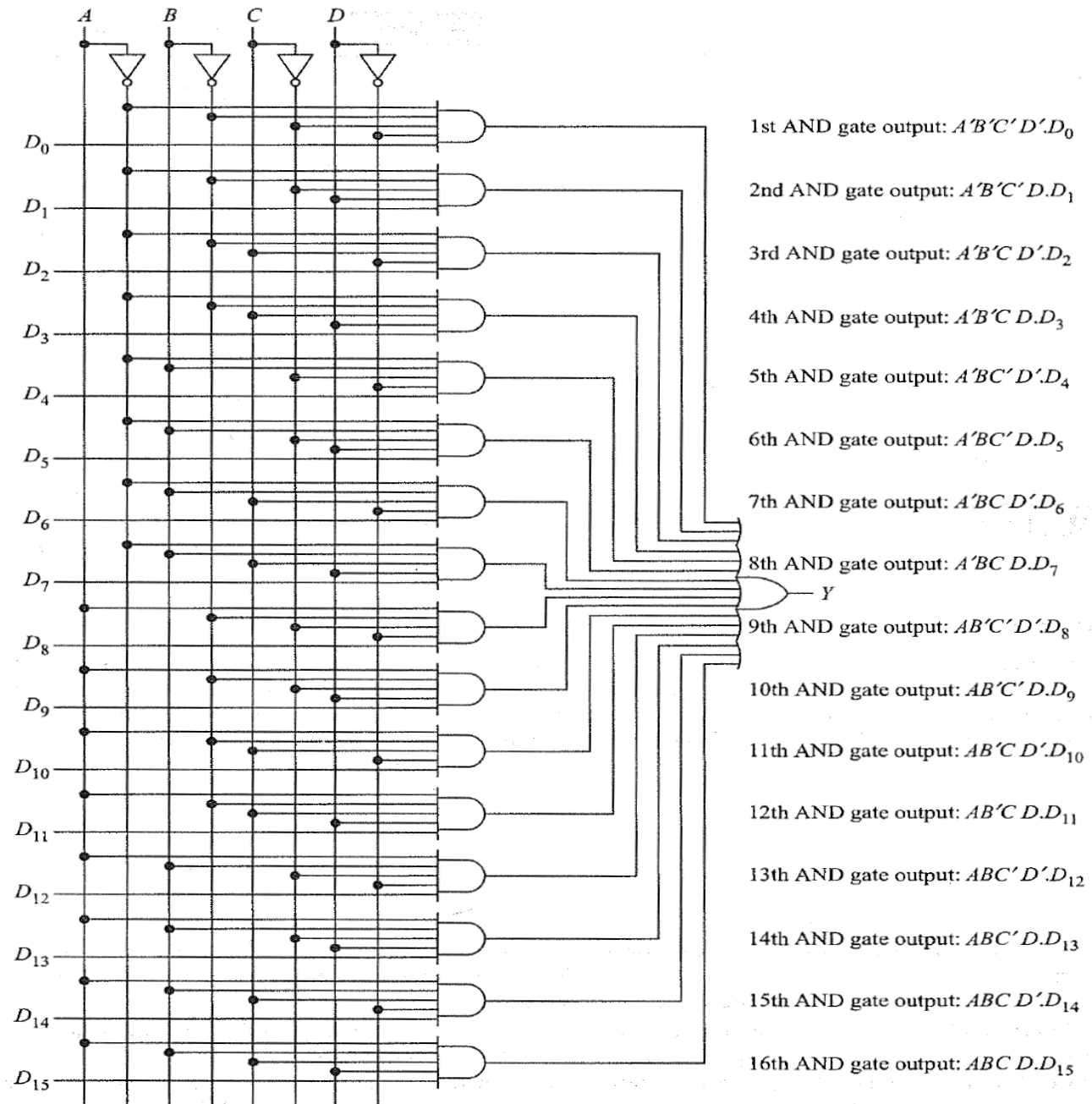


Fig. 4.1

(a) Multiplexer block diagram, (b) 4-to-1 multiplexer truth table, (c) Its logic circuit

- 16:1 Multiplexer



16:1 Multiplexer (IC 74150)

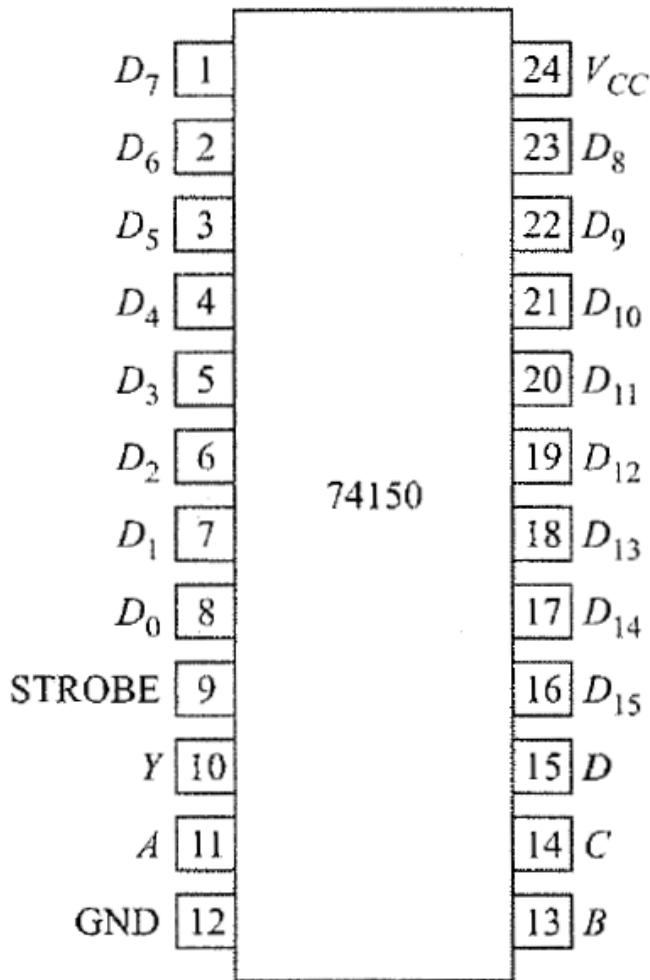


Table 4.1

74150 Truth Table

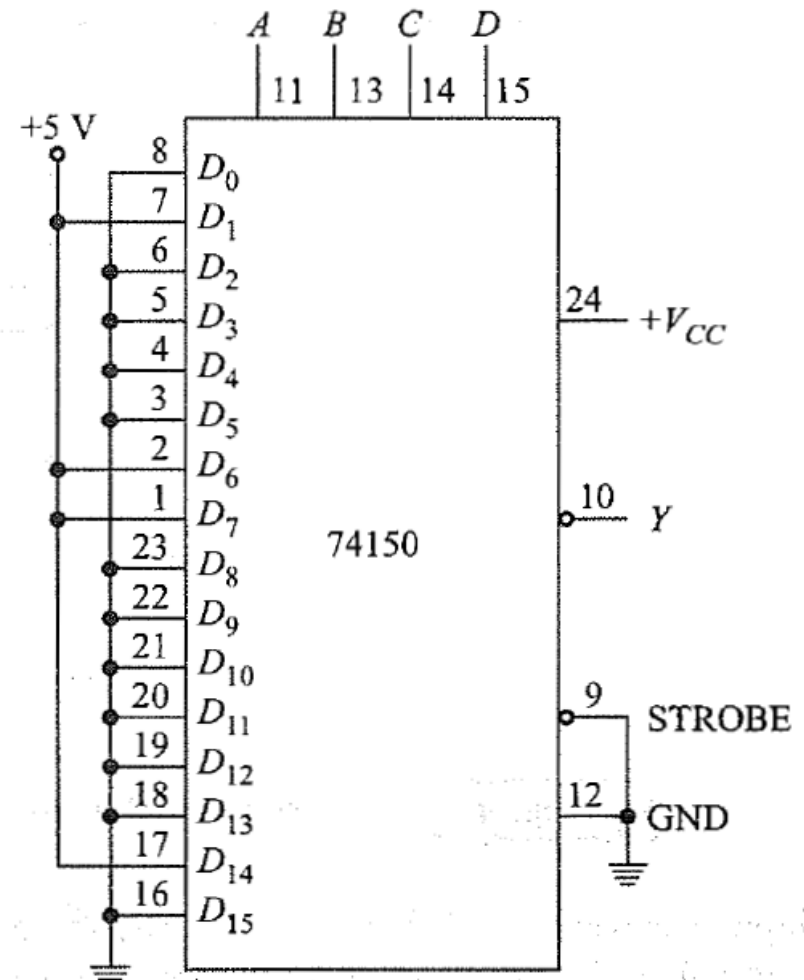
Strobe	A	B	C	D	Y
L	L	L	L	L	$\overline{D_0}$
L	L	L	L	H	$\overline{D_1}$
L	L	L	H	L	$\overline{D_2}$
L	L	L	H	H	$\overline{D_3}$
L	L	H	L	L	$\overline{D_4}$
L	L	H	L	H	$\overline{D_5}$
L	L	H	H	L	$\overline{D_6}$
L	L	H	H	H	$\overline{D_7}$
L	H	L	L	L	$\overline{D_8}$
L	H	L	L	H	$\overline{D_9}$
L	H	L	H	L	$\overline{D_{10}}$
L	H	L	H	H	$\overline{D_{11}}$
L	H	H	L	L	$\overline{D_{12}}$
L	H	H	L	H	$\overline{D_{13}}$
L	H	H	H	L	$\overline{D_{14}}$
L	H	H	H	H	$\overline{D_{15}}$
H	X	X	X	X	H

Fig. 4.3 Pinout diagram of 74150

Realization of Truth table using Mux

Table 4.2

A	B	C	D	Y
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	0
1	1	1	1	1

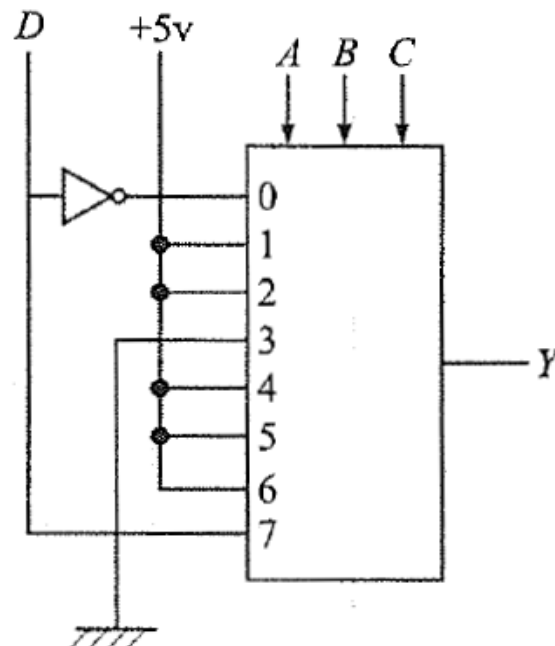


complement of selected data bit

4 variable Truth Table realization using 8:1 Mux

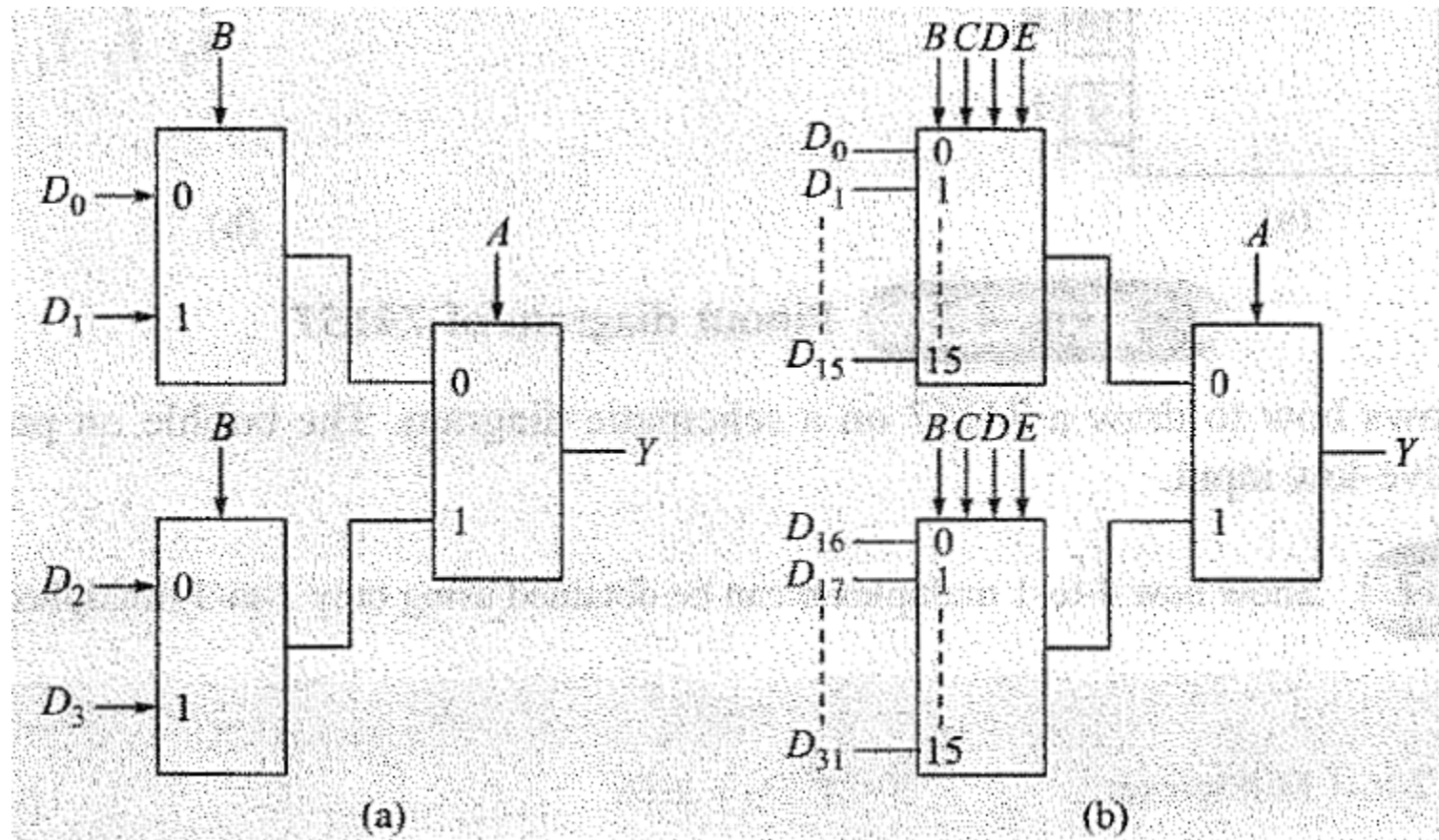
ABC	000	001	010	011	100	101	110	111
$D = 0$	1	1	1	0	1	1	1	0
$D = 1$	0	1	1	0	1	1	1	1
Y	D'	1	1	0	1	1	1	D
8-to-1 MUX data input	$D_0 = D'$	$D_1 = 1$	$D_2 = 1$	$D_3 = 0$	$D_4 = 1$	$D_5 = 1$	$D_6 = 1$	$D_7 = D$

(a)

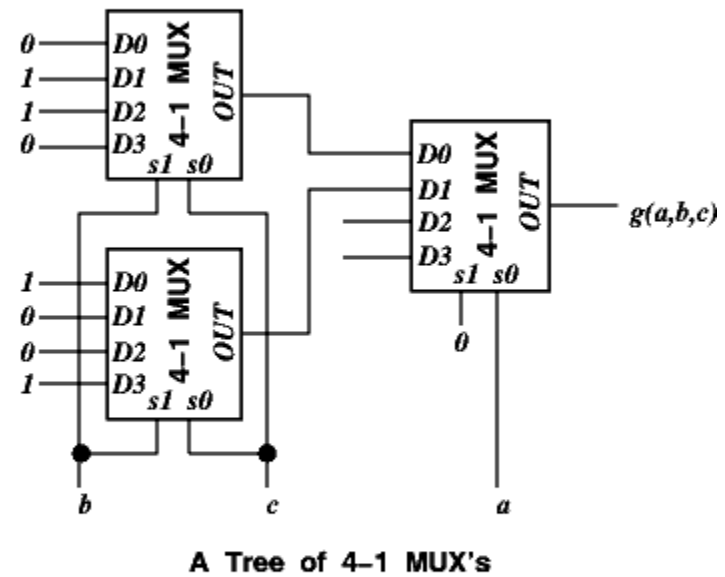
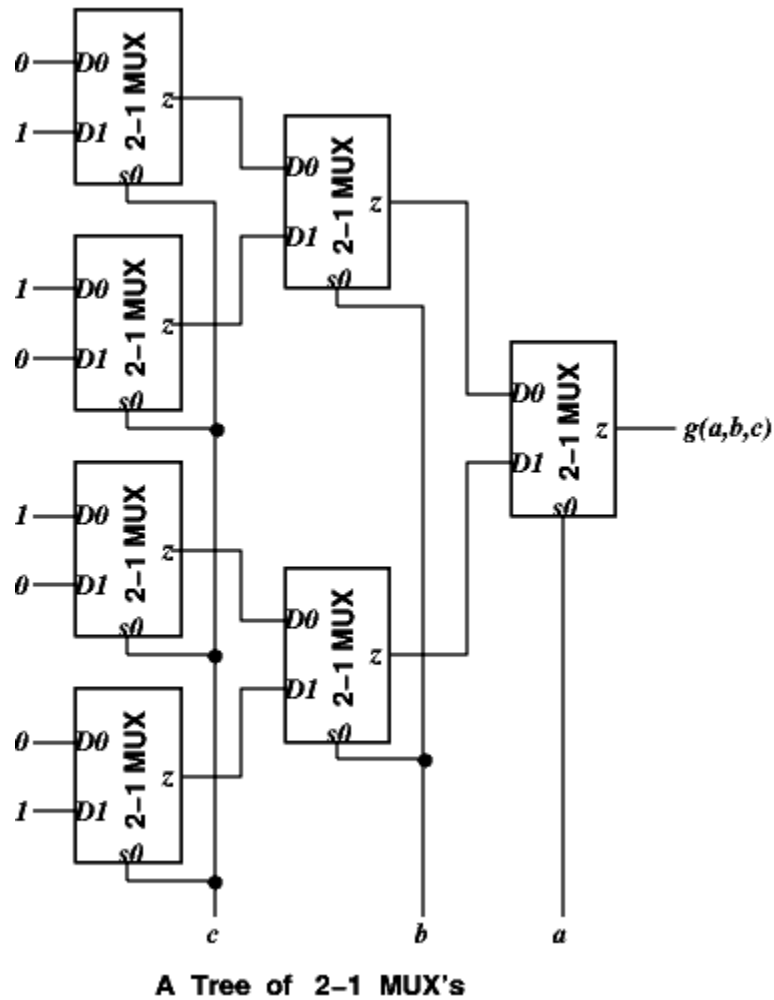


(b)

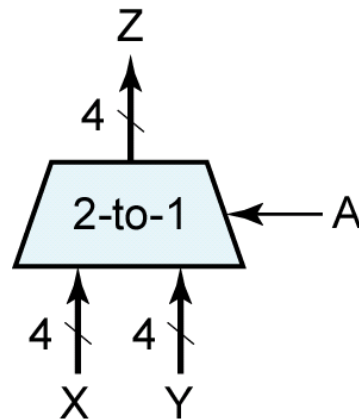
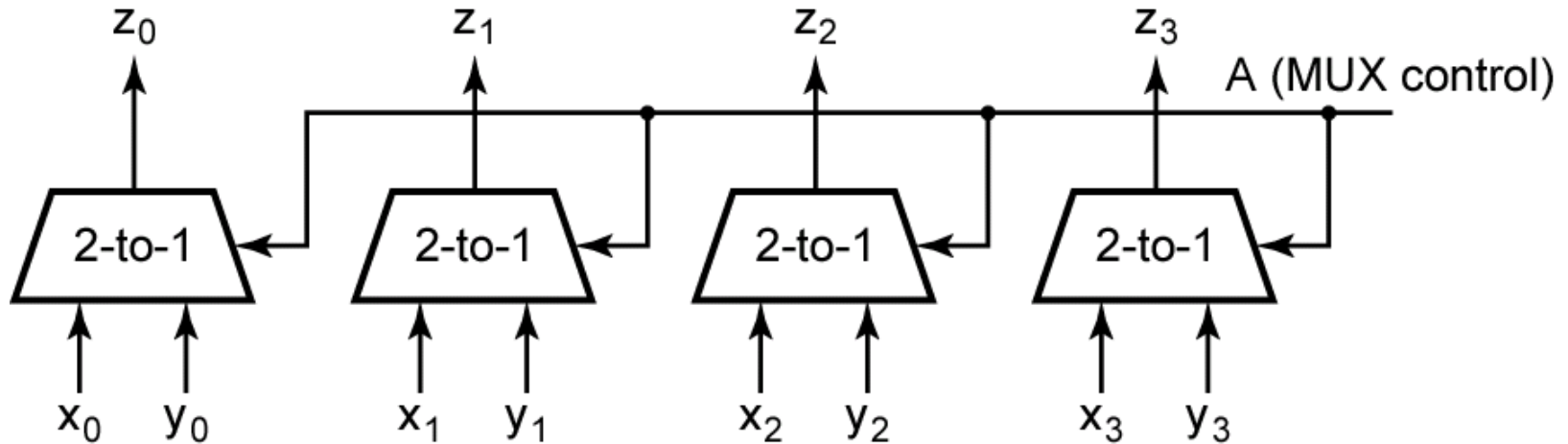
Multiplexer Tree



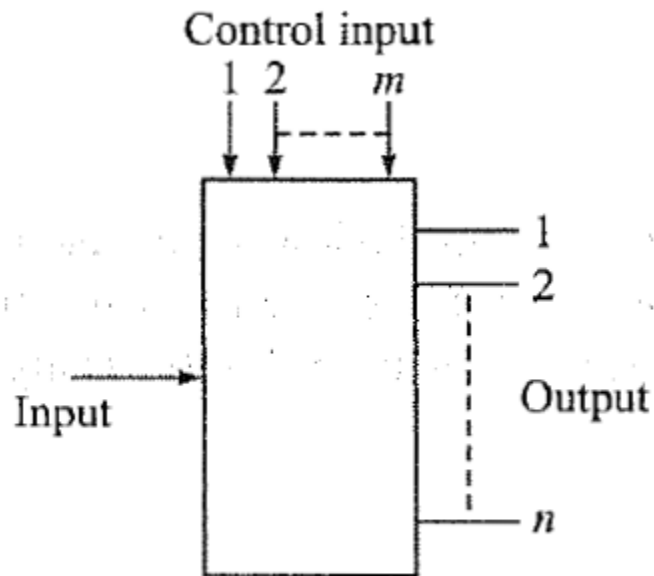
Multiplexer Tree: A Larger Multiplexer can be implemented using smaller Multiplexers



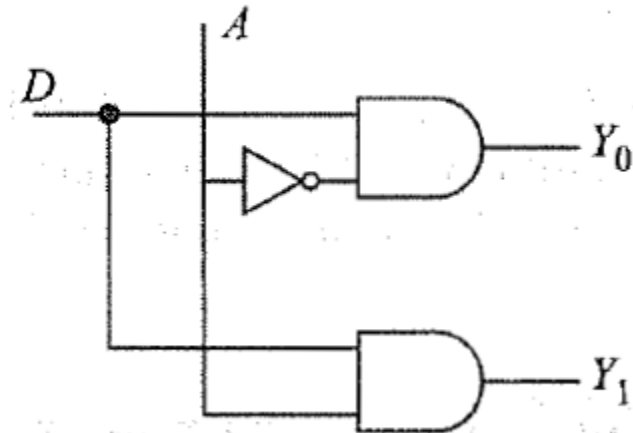
Nibble Multiplexer/ Multiplexer Bus



Demultiplexer



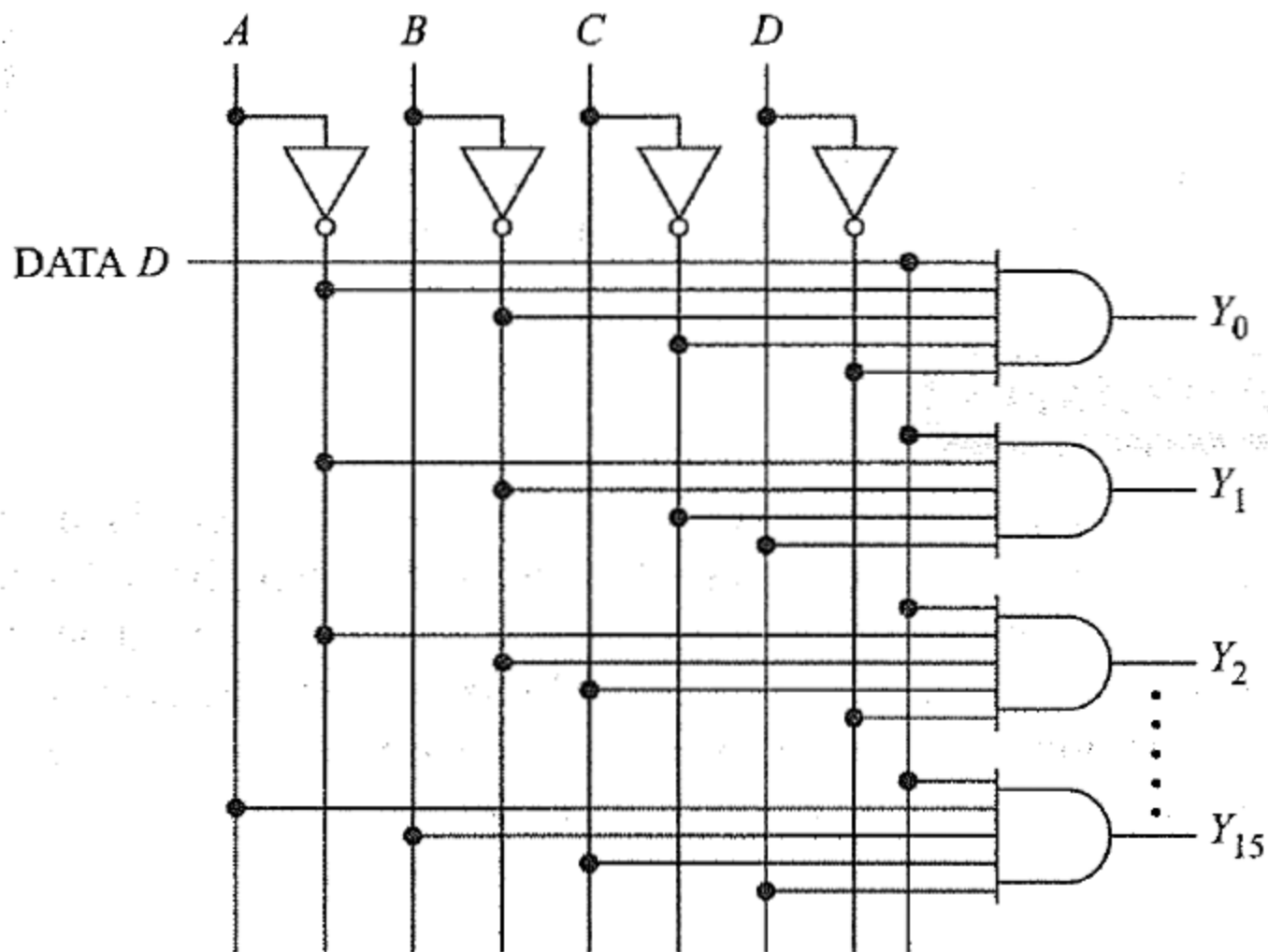
(a)



(b)

(a) Demultiplexer block diagram, (b) Logic circuit of 1-to-2 demultiplexer

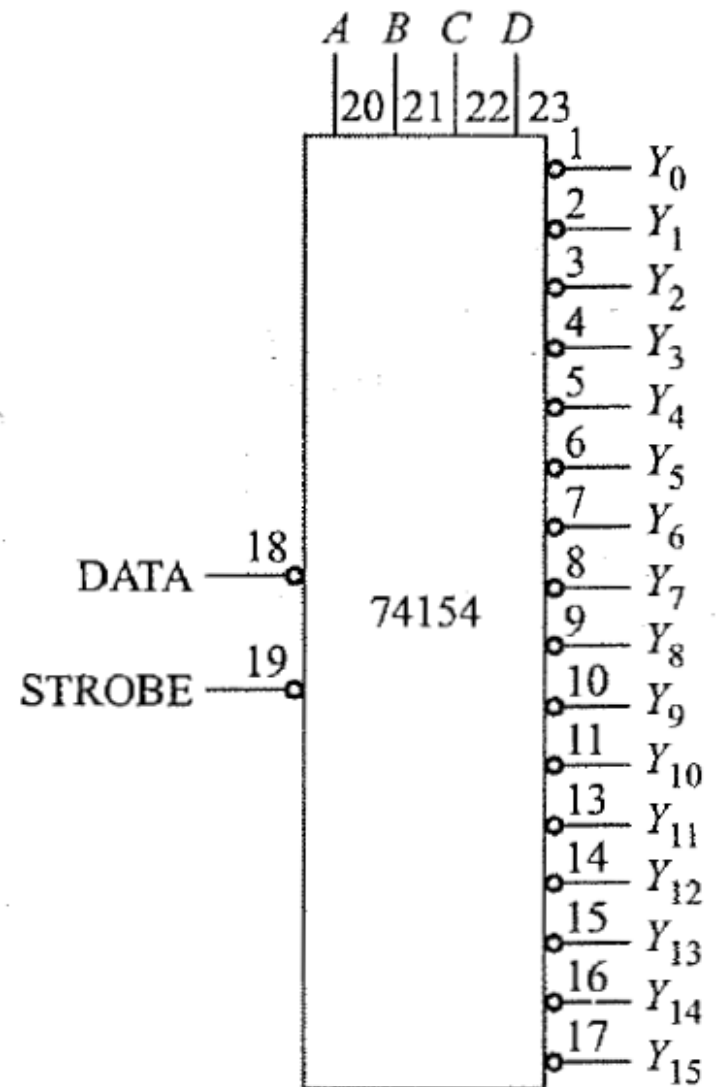
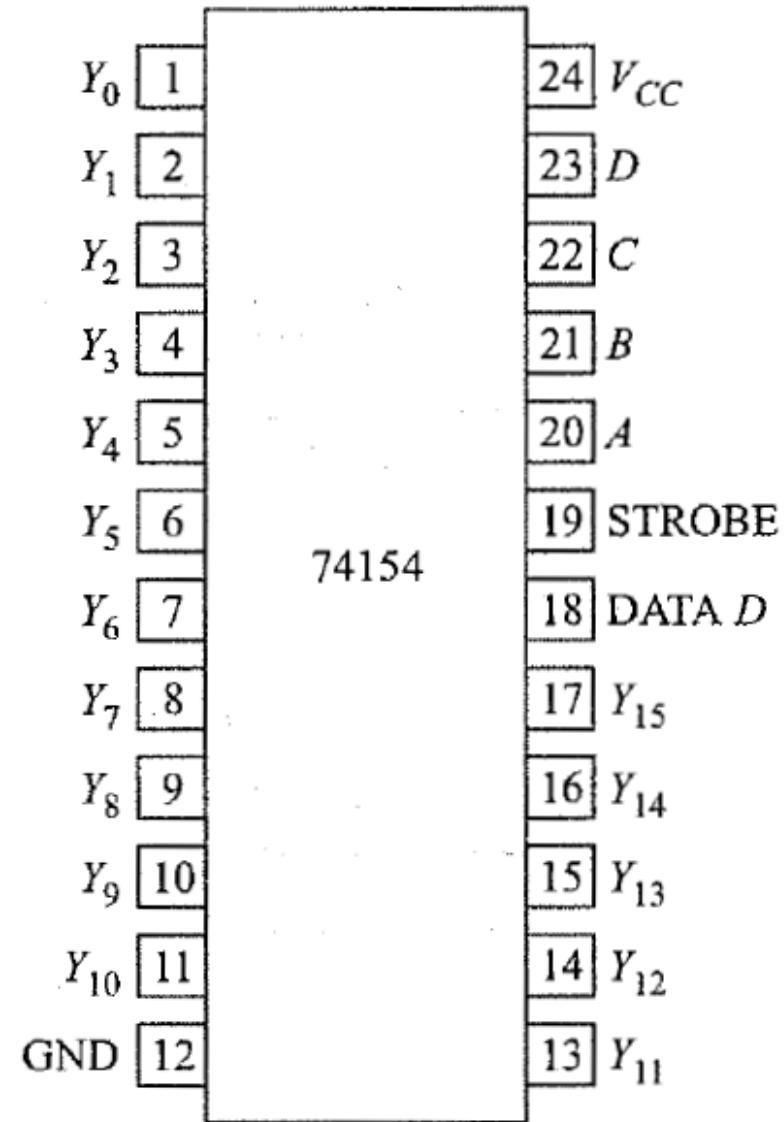
1-to-16 demultiplexer



IC No.	DEMUX Type	Decoder Type
74154	1-to-16	4-to-16
74138	1-to-8	3-to-8
74155	1-to-4	2-to-4

74154 (1:16 Demux)

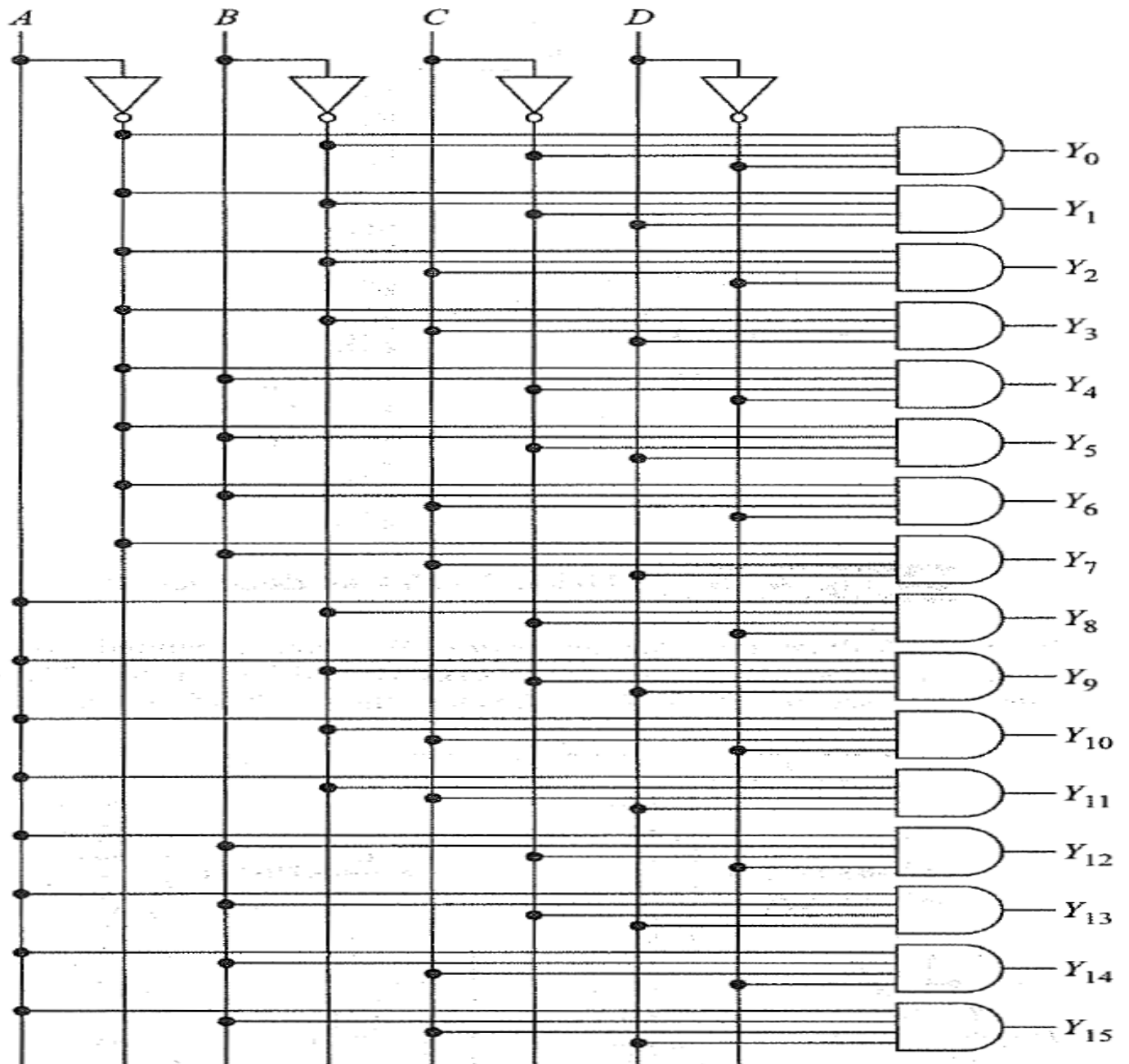
Pin out Diagram and Logic Diagram



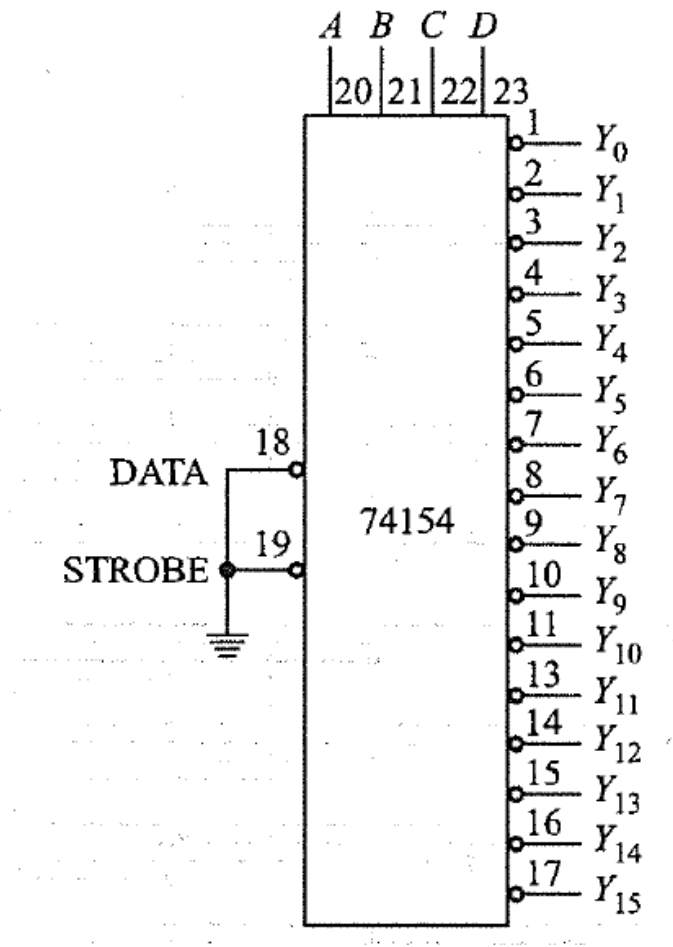
74154 Truth Table(1:16 Demux)

[illegible]

IC 74154
as 4:16
Decoder



IC 74154 as 4:16 Decoder Logic Diagram



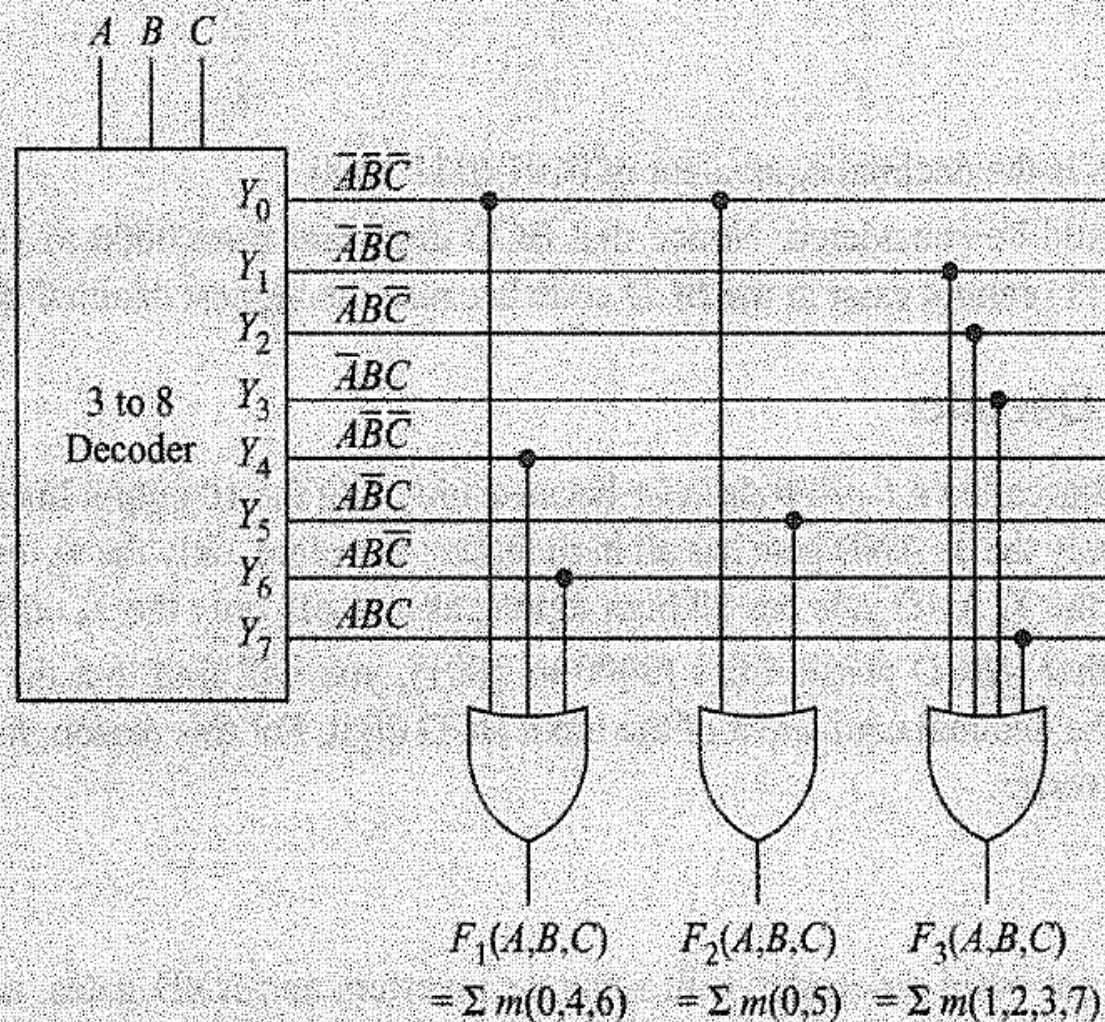


Example 4.7

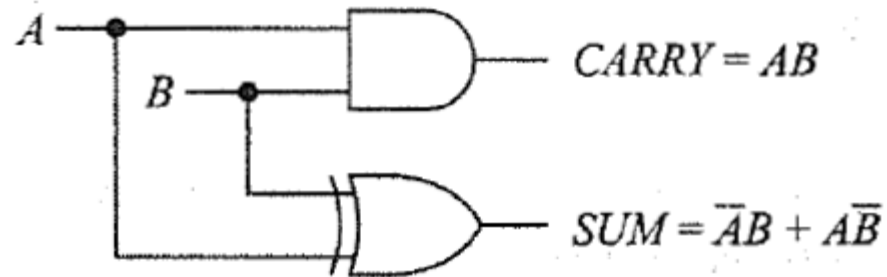
Show how using a 3-to-8 decoder and multi-input OR gates following Boolean expressions can be realized simultaneously.

$$F_1(A, B, C) = \Sigma m(0, 4, 6); F_2(A, B, C) = \Sigma m(0, 5); F_3(A, B, C) = \Sigma m(1, 2, 3, 7)$$

Solution Since at the decoder output we get all the minterms we use them as shown in Fig. 4.17 to get the required Boolean functions.



Half Adder



A	B	$CARRY$	SUM
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

Full Adder

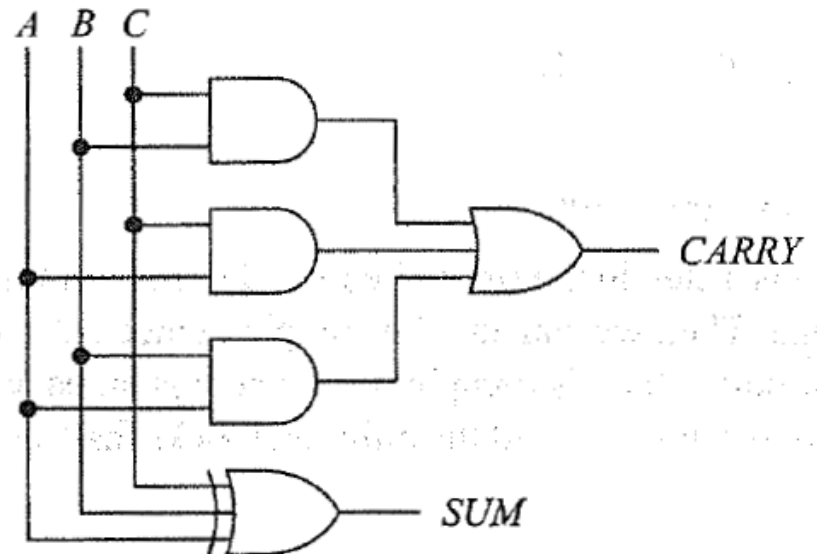
<i>A</i>	<i>B</i>	<i>C</i>	<i>CARRY</i>	<i>SUM</i>
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

<i>A B</i>	00	01	11	10
<i>C</i> 0	0	0	1	0
1	0	1	1	1

$$\text{Carry} = AB + BC + AC$$

<i>A B</i>	00	01	11	10
<i>C</i> 0	0	1	0	1
1	1	0	1	0

$$\text{SUM} = A \oplus B \oplus C$$



Binary Subtractor

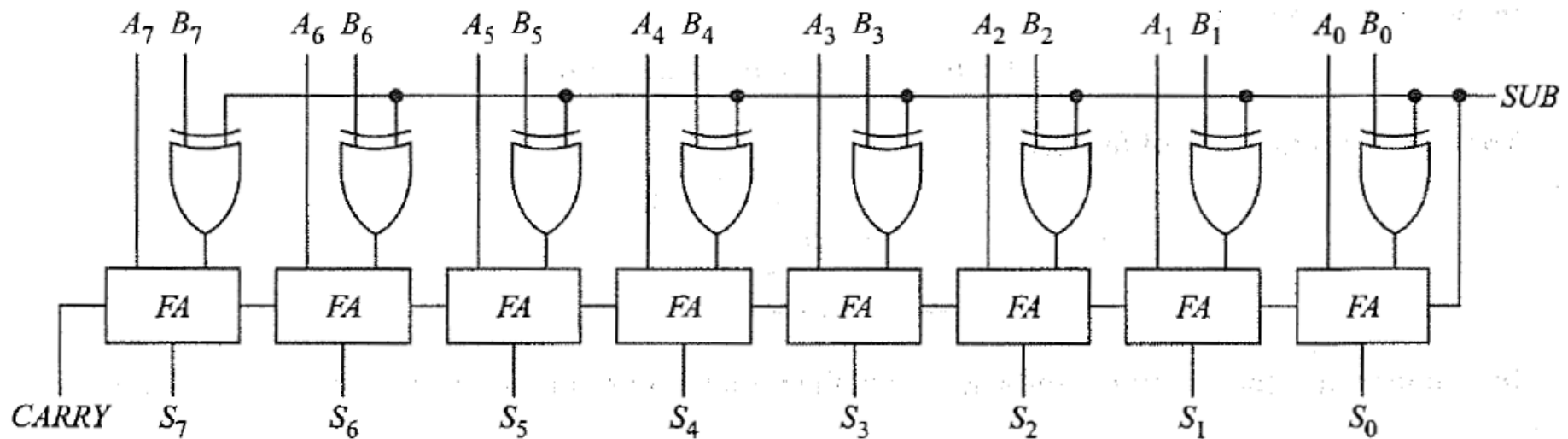
- Basic Binary Subtraction Rules

$$0 - 0 = 0$$

$$1 - 0 = 1$$

$$1 - 1 = 0$$

$$10 - 1 = 1$$



Reference:

- Book: DIGITAL PRINCIPLES AND APPLICATION BY LEACH & MALVINO (Publisher: Tata Mc-Graw Hill Education private Limited)